

Practice Test III (Version 1)

Date
Course

- I) Write your name, the date, and the course title (i.e. Precalculus).
- II) Show all work.
- III) Draw a rectangle around your final answer for each problem.
- IV) A graphing calculator is required, but is not allowed for some problems.

Convert the rectangular coordinates below to polar coordinates. Use degree units for angle measures. Round to the nearest hundredth if necessary.

① $(1, 4)$ $(x, y) \rightarrow (r, \theta)$

I) Find r .

$$x^2 + y^2 = r^2$$

$$1^2 + 4^2 = r^2$$

$$r = \sqrt{17}$$

II) Find θ

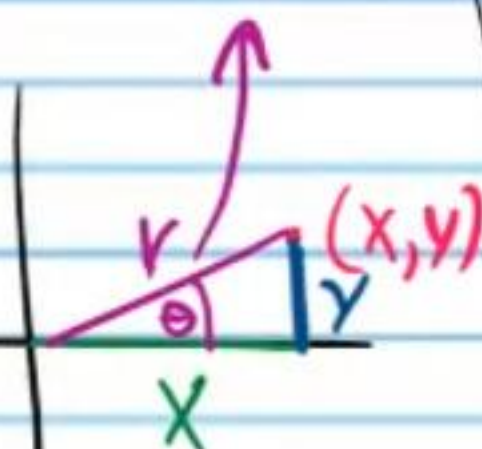
$$\tan \theta = \frac{y}{x}$$

$$\tan \theta = \frac{4}{1}$$

$$\theta = \tan^{-1} 4$$

$$\theta \approx 75.96^\circ$$

$$\boxed{(\sqrt{17}, 75.96^\circ)}$$



② $(-3, -5)$

I) Find r .

$$x^2 + y^2 = r^2$$

$$(-3)^2 + (-5)^2 = r^2$$

$$r = \sqrt{34}$$

II) Find θ .

$$\tan \theta = \frac{y}{x}$$

$$\tan \theta = \frac{-5}{-3}$$

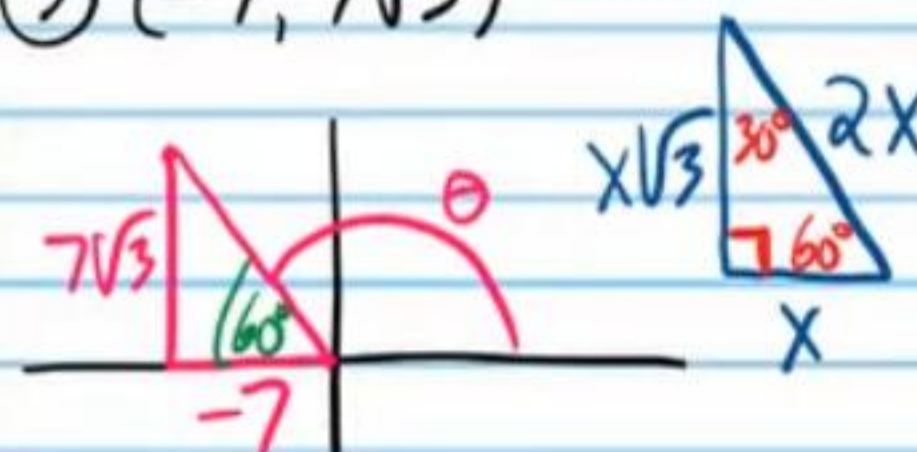
$$\theta = \tan^{-1} \left(\frac{5}{3} \right)$$

$$\theta \approx 59.03624^\circ$$

$$\boxed{(-\sqrt{34}, 59.04^\circ)}$$

* You may not use a calculator.

③ $(-7, 7\sqrt{3})$

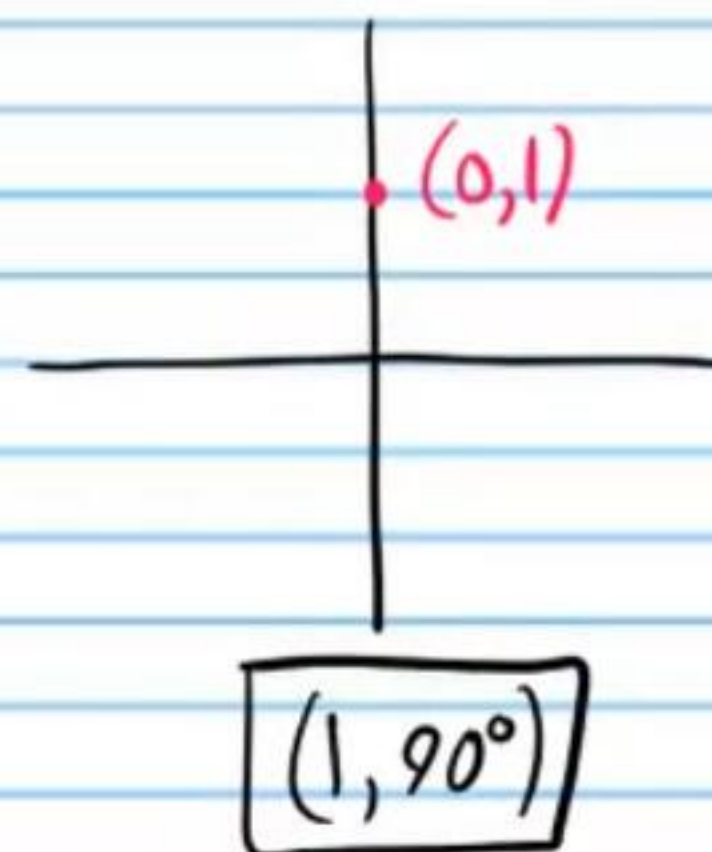


$$\theta = 120^\circ$$

$$r = 2x = 2(7) = 14$$

$$\boxed{(14, 120^\circ)}$$

④ $(0, 1)$



Convert the following polar coordinates to rectangular coordinates.
Round to the nearest thousandth if necessary.

⑤ $(-2, -308^\circ)$

Find x .

$$x = r \cos \theta$$

$$x = -2 \cos(-308^\circ)$$

$$x \approx -1.231$$

Find y .

$$y = r \sin \theta$$

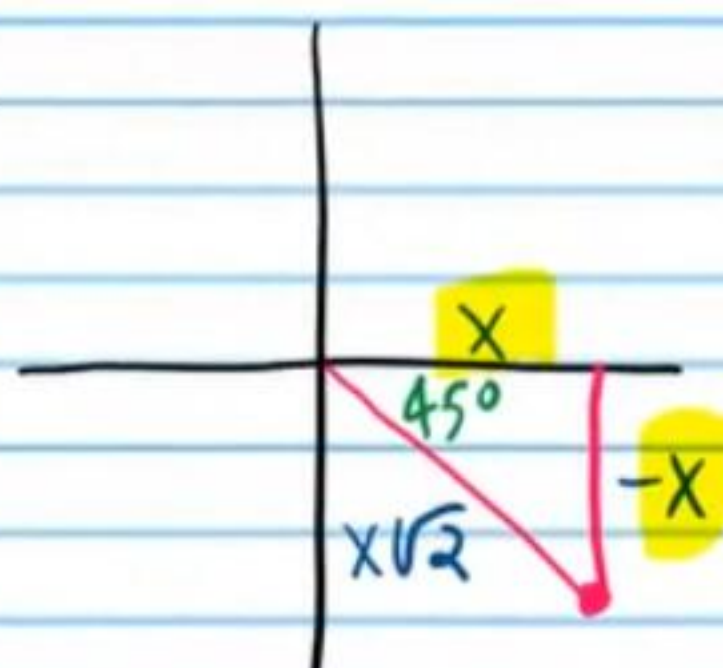
$$y = -2 \sin(-308^\circ)$$

$$y \approx -1.576$$

$$\boxed{(-1.231, -1.576)}$$

⑥ $(-3, 135^\circ)$

You may not use a calculator.



$$x\sqrt{2} = 3$$

$$x = \frac{3}{\sqrt{2}}$$

$$x = \frac{3\sqrt{2}}{2}$$

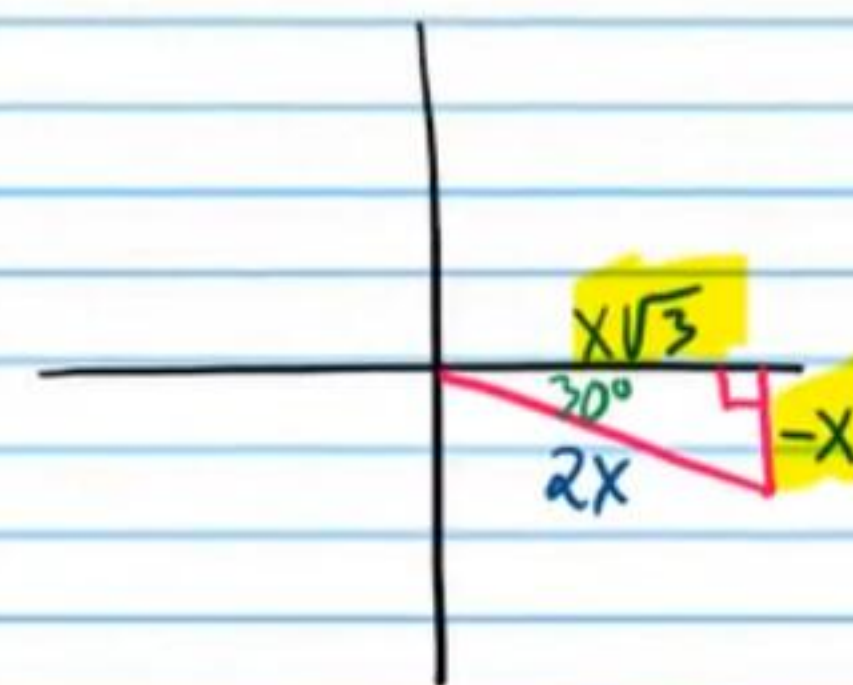
$$\boxed{\left(\frac{3\sqrt{2}}{2}, -\frac{3\sqrt{2}}{2}\right)}$$

You may not use a calculator.

⑦ $(0, 50^\circ)$

$$\boxed{(0, 0)}$$

⑧ $(12, -30^\circ)$



$$2x = 12$$

$$x = 6$$

$$\boxed{(6\sqrt{3}, -6)}$$

⑨ A relation is written below in table form. Perform the following reflections and fill in the tables.

X	10	9	5	0	-2
Y	8	81	-90	39	46

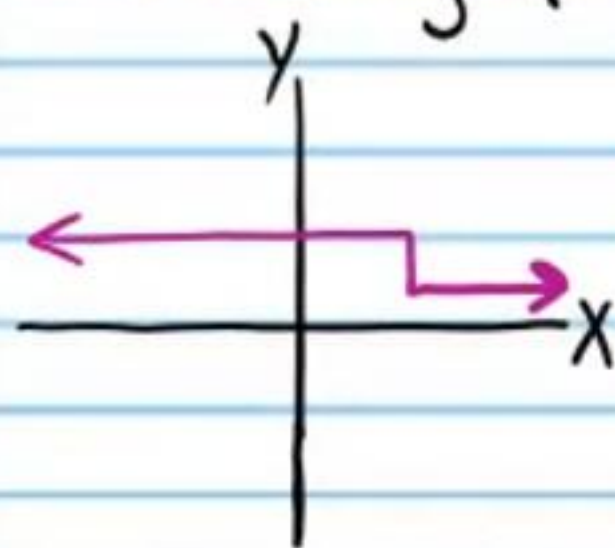
i) **Reflect across the x-axis** →

X	10	9	5	0	-2
Y	-8	-81	90	-39	-46

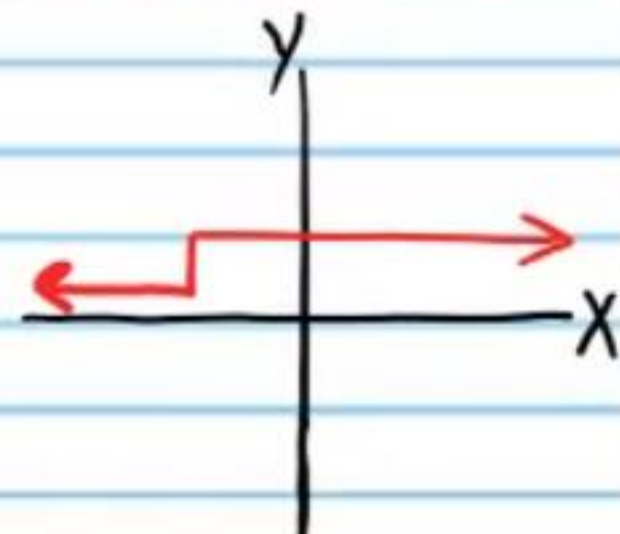
ii) **Reflect across the origin** →

X	-10	-9	-5	0	2
Y	-8	-81	90	-39	-46

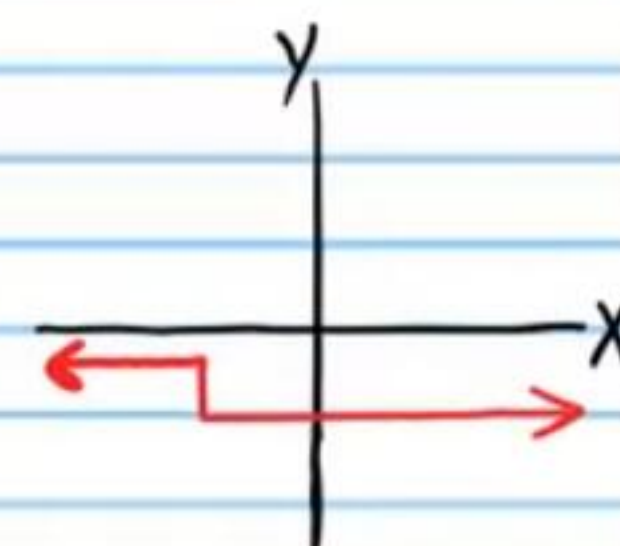
A relation is written below in the form of a graph. Draw the graphs of the indicated reflections.



iii) **Reflect across the y-axis** →



iv) **Reflect across the origin** →



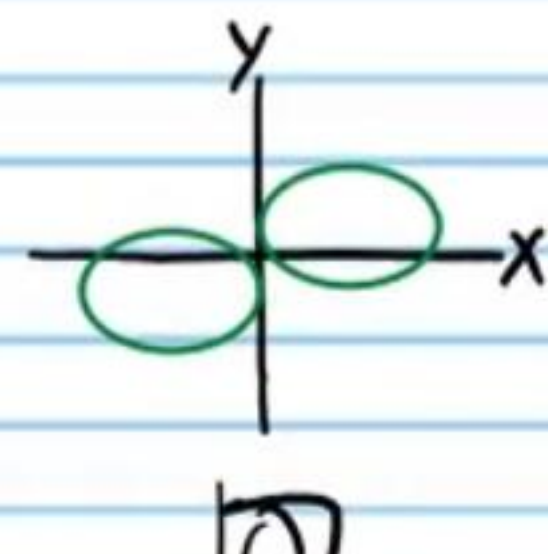
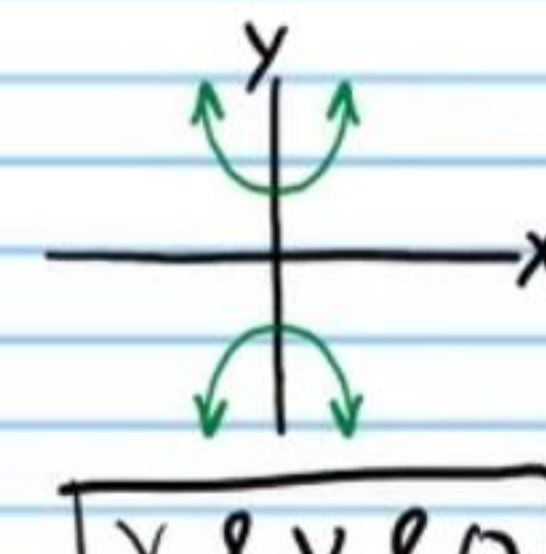
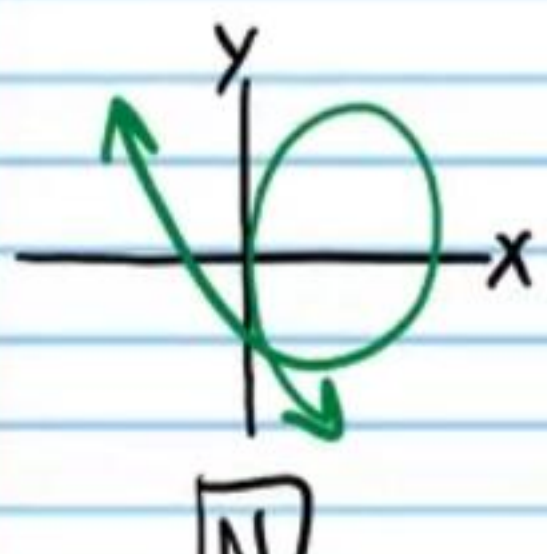
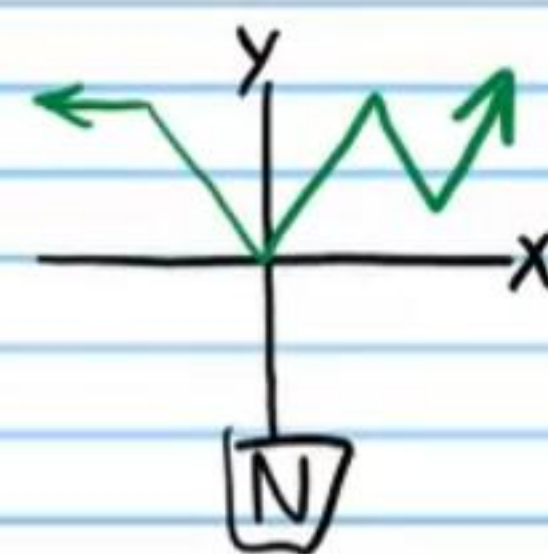
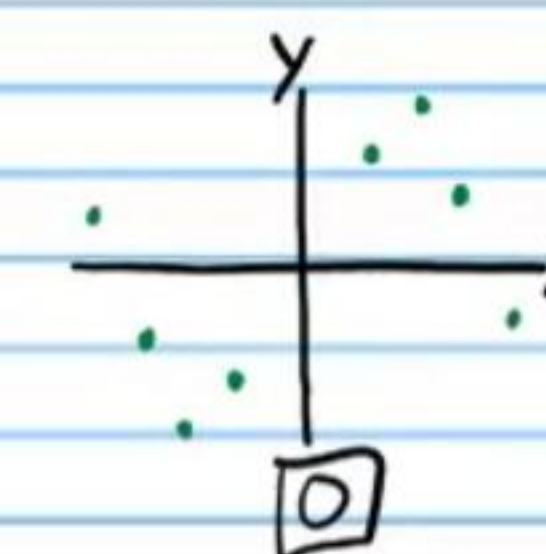
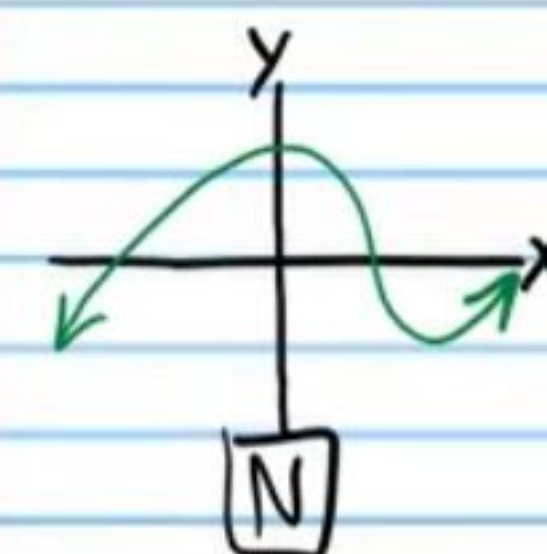
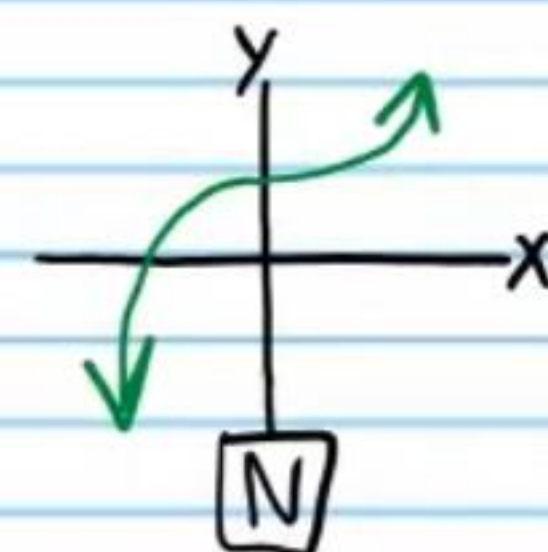
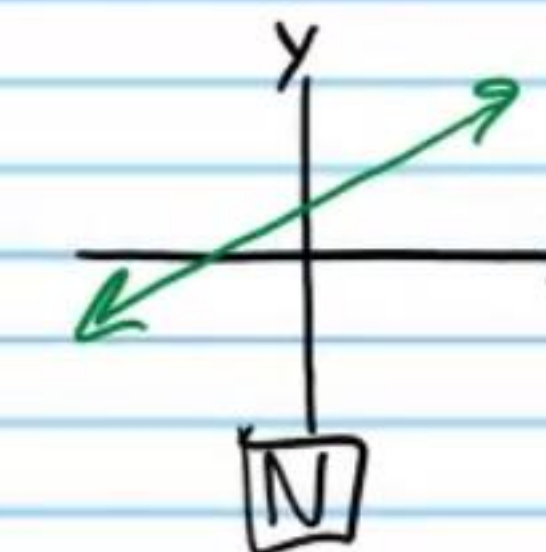
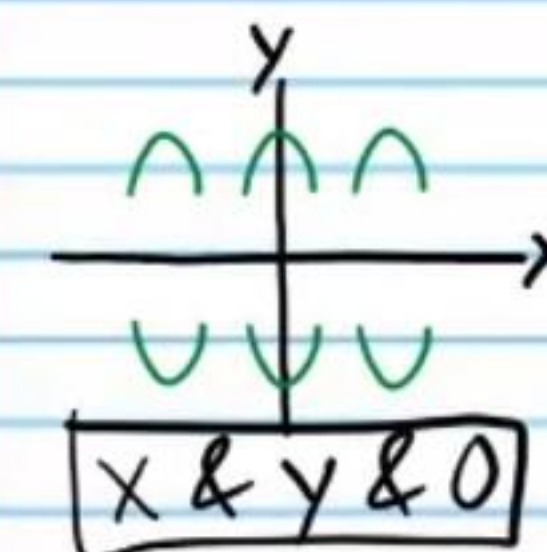
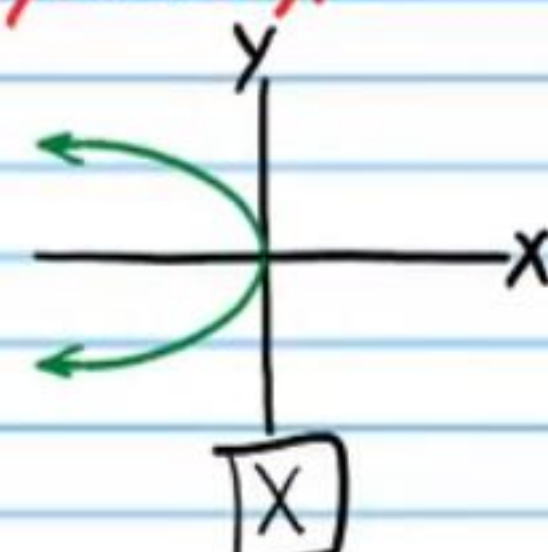
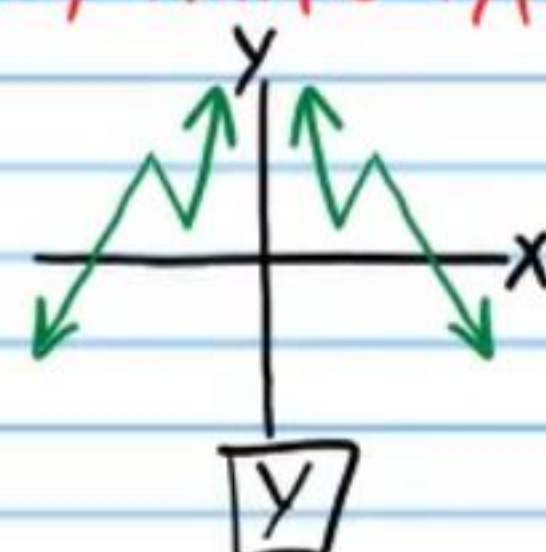
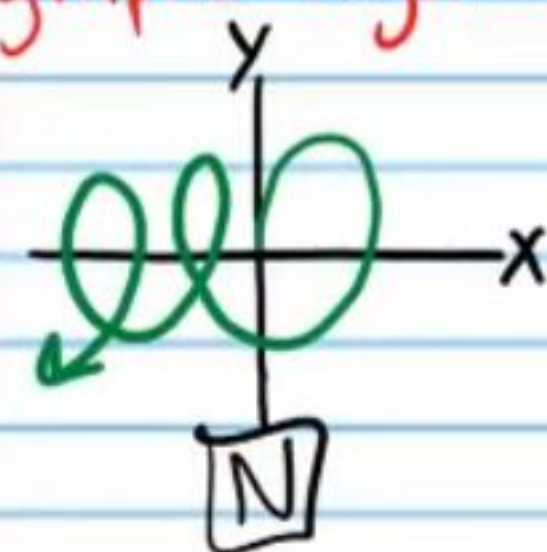
⑩ i) If "f" is an even function, and $f(-1) = -2$, find $f(1)$.

$$(x, y) \rightarrow (-x, y) \quad \boxed{f(1) = -2}$$

ii) If "f" is an odd function, and $f(\frac{1}{2}) = \frac{3}{7}$, find $f(-\frac{1}{2})$.

$$(x, y) \rightarrow (-x, -y) \quad \boxed{f(-\frac{1}{2}) = -\frac{3}{7}}$$

⑪ Relations are written below in the form of graphs. If a relation appears to be symmetric about the x-axis, write "x". If a relation appears to be symmetric about the y-axis, write "y". If a relation appears to be symmetric about the origin, write "O". If the relation shows no apparent symmetry, write "N". Some graphs might display multiple types of symmetry.



⑫ Each table represents a relation that has the indicated symmetry. Fill in the missing parts of the table.

i) Symmetric about the X-axis.

X	13	0	-8	7	4	13	0	-8	7	4
Y	10	3	16	-10	20	-10	-3	-16	10	-20

ii) Symmetric about the Y-axis.

X	1	-2	14	67	30	-1	2	-14	-67	-30
Y	-1	4	-17	49	-20	-1	4	-17	49	-20

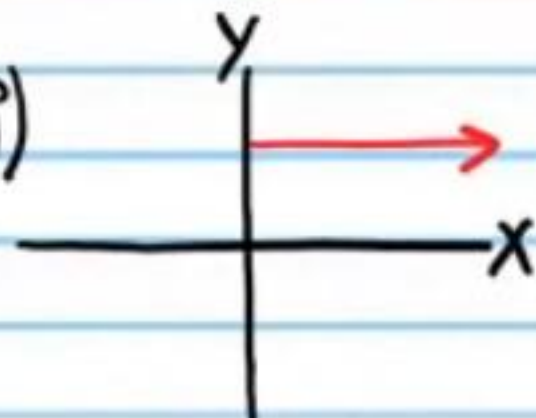
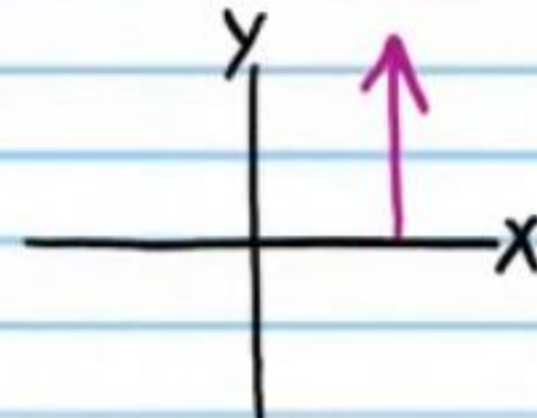
⑬ Relations are written below in various forms. Write each inverse relation in its corresponding form.

i)

X	-2	-8	-7	0	2
Y	-3	-1	0	4	5

 Write the inverse relation
i.e. Reflection across $y=x$

X	-3	-1	0	4	5
Y	-2	-8	-7	0	2

ii)  Write the inverse relation
i.e. Reflection across $y=x$ 

iii) $\tan x - 1 = y$ Write the inverse relation
i.e. Reflection across $y=x$ $\tan y - 1 = x$

iv) $y + x^2 = 2x - y^2$ Write the inverse relation
i.e. Reflection across $y=x$ $x + y^2 = 2y - x^2$

⑭ i) Create parametric equations for the relation below. Use "t" as the parameter.

$$y = \frac{1}{x}$$

$x = t, y = \frac{1}{t}$

ii) Determine if the function below is even, odd, or neither.

$$f(x) = \tan x + 2x^2$$

$$f(-x) = \tan(-x) + 2(-x)^2$$

$$f(-x) = -\tan x + 2x^2$$

Neither

⑮ The equation of a relation is written below. Write the equation of each reflection. Simplify if possible.

$$y + |x| = 3y^2$$

i) Reflect across the x-axis → $-y + |x| = 3(-y)^2$
 $-y + |x| = 3y^2$

ii) Reflect across the y-axis → $y + |-x| = 3y^2$
 $y + |x| = 3y^2$

Is the relation symmetric about the

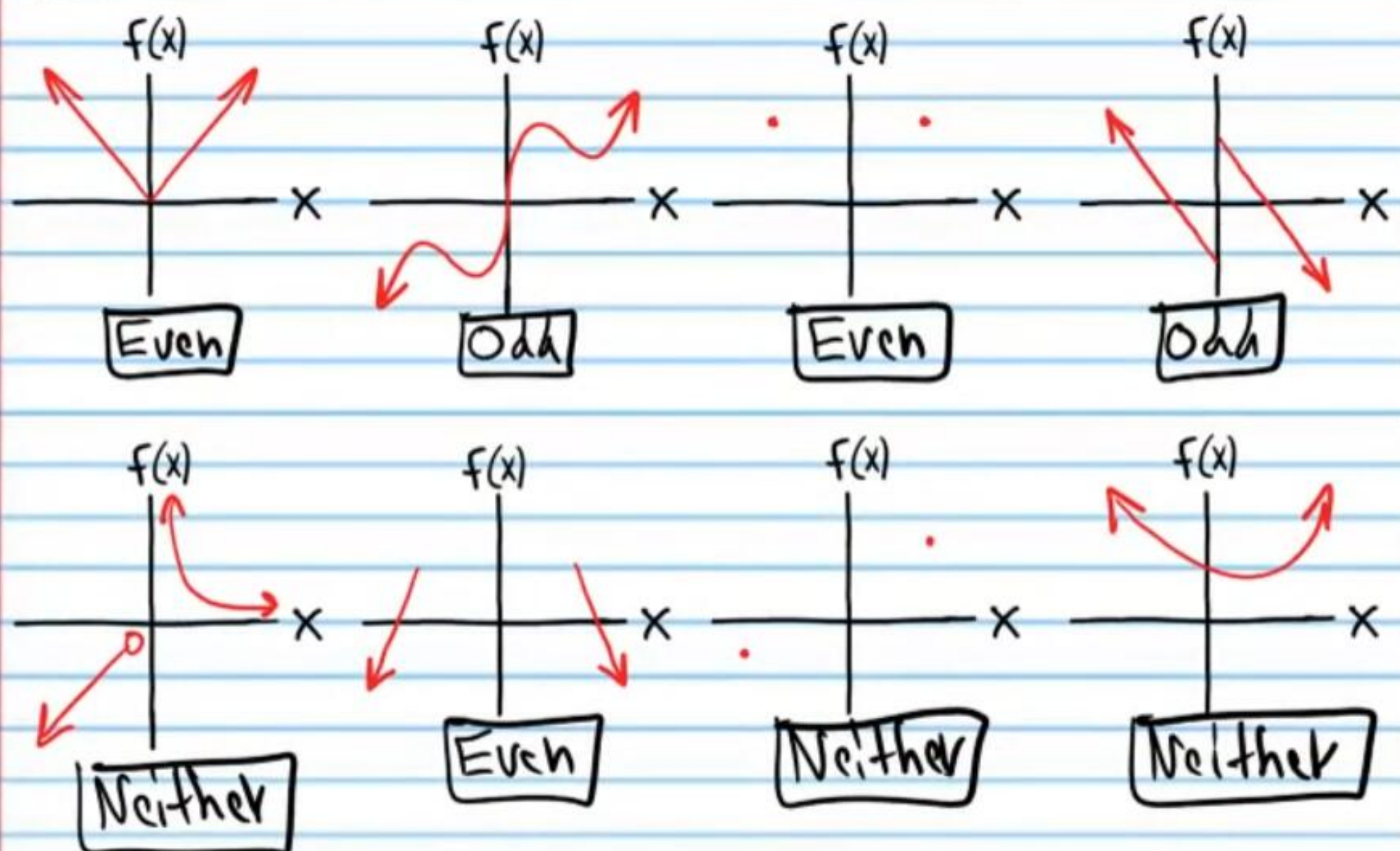
iii) x-axis? **No**

iv) y-axis? **Yes**

16) i) "f" and "g" are functions represented by the tables below. If "f" is an even function and "g" is an odd function, fill the empty spaces with other values of the functions.

X	100	40	-9	-11	7	-100	-40	9	11	-7	X	2	19	-1	-3	7	-2	-19	1	3	-7
f(x)	20	30	17	-12	7	20	30	17	-12	7	g(x)	4	5	9	0	-1	-4	-5	-9	0	1

ii) Determine if the graphs below are even functions, odd functions, or neither.



17) Solve the following equations.

i) $\frac{1}{36} = 2^x \cdot 3^x$

ii) $3\log_2 X - 5 = 7$

$\frac{1}{6^2} = 2^x \cdot 3^x$

$3\log_2 X = 12$

$6^{-2} = 2^x \cdot 3^x$

$\log_2 X = 4$

$6^{-2} = (2 \cdot 3)^x$

$2^4 = X$

$6^{-2} = 6^x$

$X = -2$

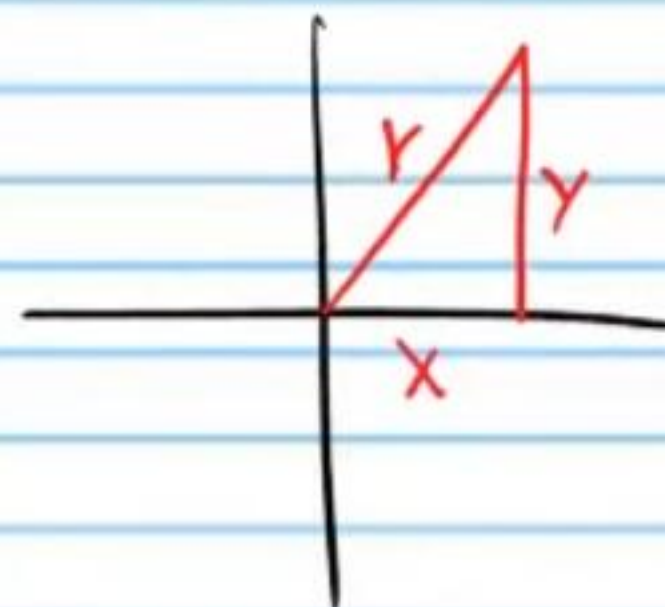
$X = 16$

18) i) Convert the equation below to rectangular form.

$r = \sqrt{3}$

$r^2 = 3$

$x^2 + y^2 = 3$



ii) Determine if the function below is even, odd, or neither.

$f(x) = \frac{x - x^3}{2}$

$f(-x) = \frac{-x - (-x)^3}{2}$

$f(-x) = \frac{-x - (-x^3)}{2}$

$f(-x) = \frac{-x + x^3}{2}$

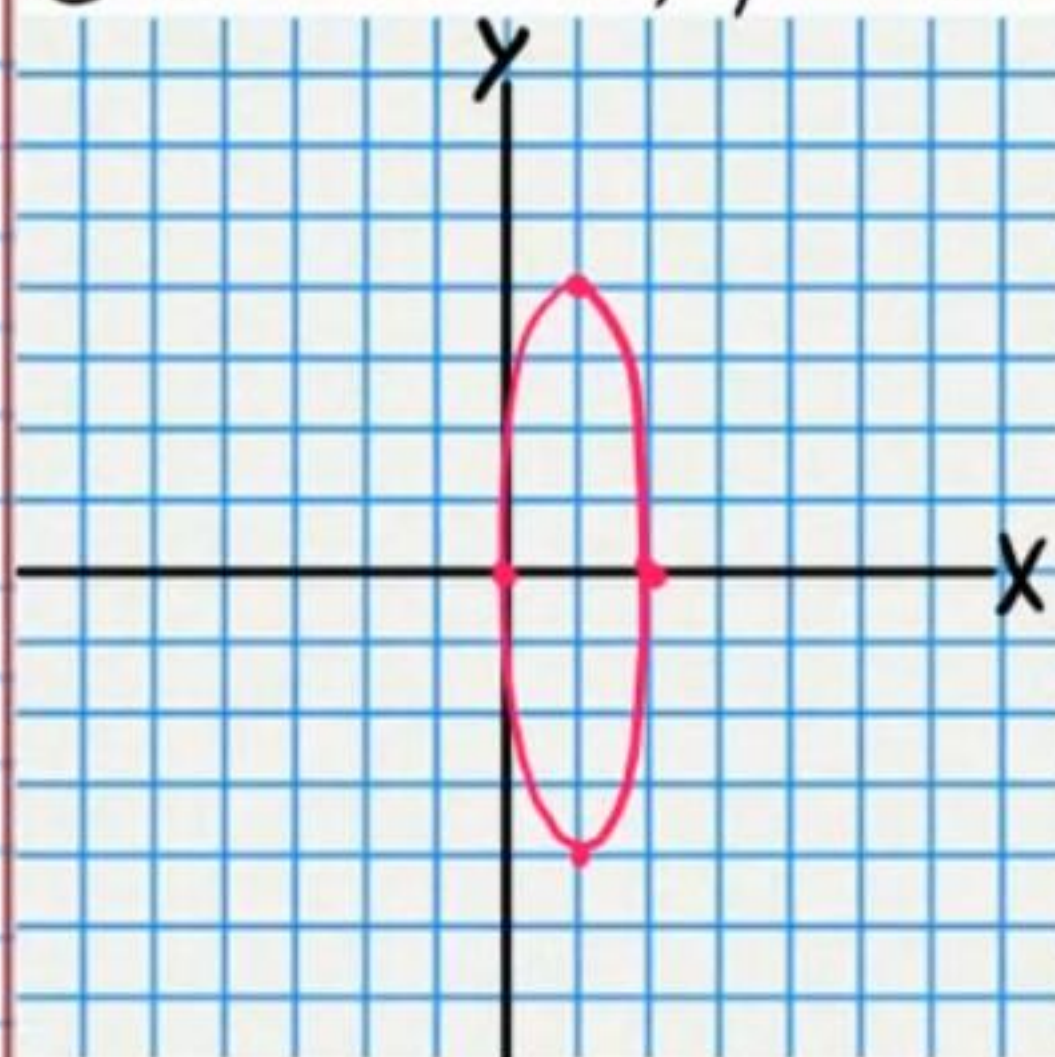
$f(-x) = -\frac{x - x^3}{2}$

$f(-x) = -f(x)$

Odd

Graph the following equations. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot. Then eliminate the parameter.

① $x = 1 + \sin t$, $y = 4 \cos t$ Point #1 Point #2



$t = 0^\circ$

$x = 1 + \sin 0^\circ$

$x = 1$

$y = 4 \cos(0)^\circ$

$y = 4(1)$

$y = 4$

$(1, 4)$

$t = 90^\circ$

$x = 1 + \sin 90^\circ$

$x = 1 + 1$

$x = 2$

$y = 4 \cos 90^\circ$

$y = 4(0)$

$y = 0$

$(2, 0)$

Point #3

$t = -90^\circ$

$x = 1 + \sin(-90^\circ)$ $y = 4 \cos(-90^\circ)$

$x = 1 - 1$ $y = 4(0)$

$x = 0$ $y = 0$

$(0, 0)$

Point #4 $x = 1 + \sin 180^\circ$, $y = 4 \cos 180^\circ$

$t = 180^\circ$ $x = 1 + 0$ $y = 4(-1)$

$(1, -4)$

Eliminate the parameter

$x = 1 + \sin t$ $y = 4 \cos t$

$\sin t = x - 1$ $\cos t = \frac{y}{4}$

$\sin^2 t + \cos^2 t = 1$

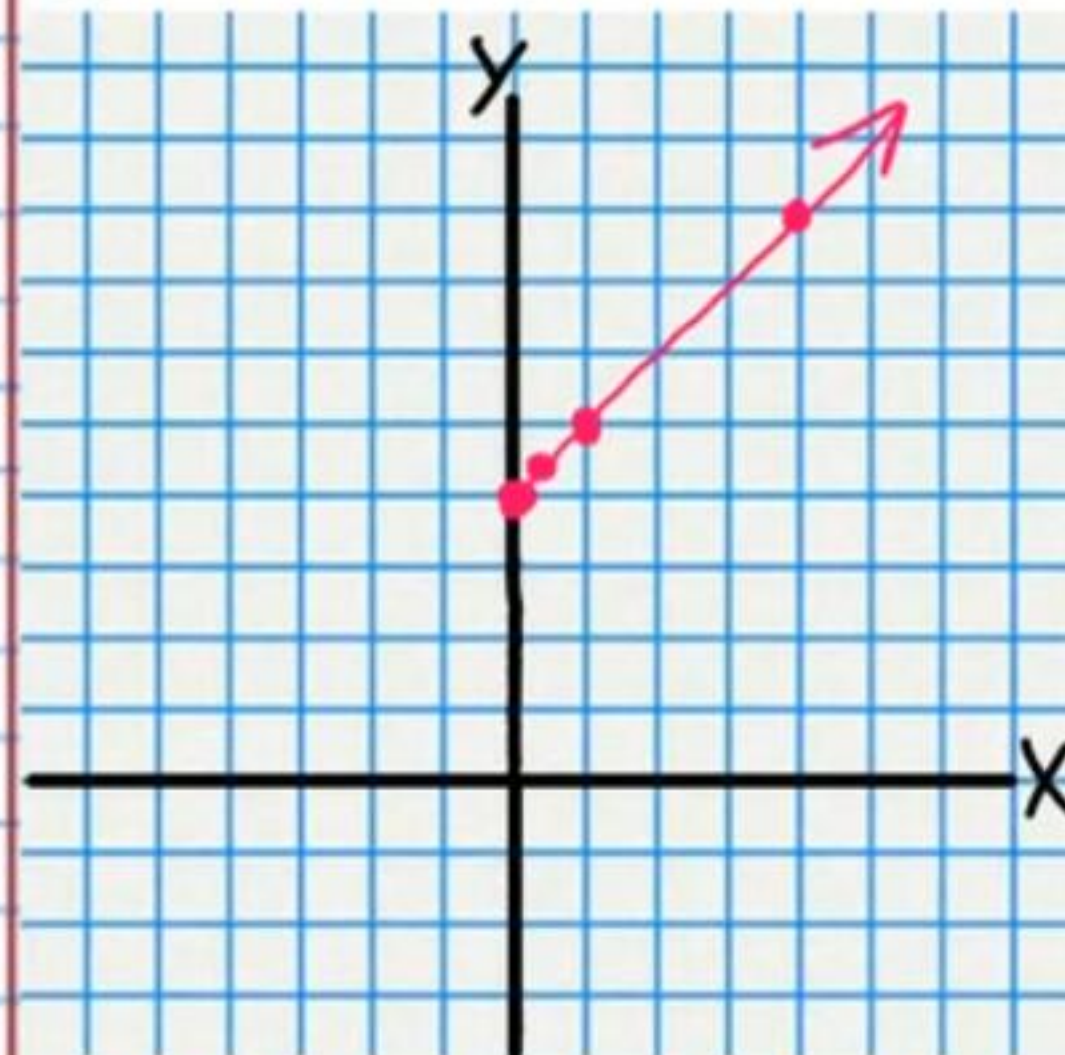
$(x - 1)^2 + \left(\frac{y}{4}\right)^2 = 1$

$(x - 1)^2 + \frac{y^2}{16} = 1$

② $x = t^2$, $y = t^2 + 4$

Point #1

Point #2



$t = 0$

$x = 0^2$

$x = 0$

$y = 0^2 + 4$

$y = 4$

$(0, 4)$

$t = 1$

$x = 1^2$

$x = 1$

$y = 1^2 + 4$

$y = 5$

$(1, 5)$

Point #3

$t = 2$

$x = 2^2$ $y = 2^2 + 4$

$x = 4$ $y = 8$

$(4, 8)$

Point #4

$t = \frac{1}{2}$

$x = \left(\frac{1}{2}\right)^2$ $y = \left(\frac{1}{2}\right)^2 + 4$

$x = \frac{1}{4}$ $y = 4\frac{1}{4}$

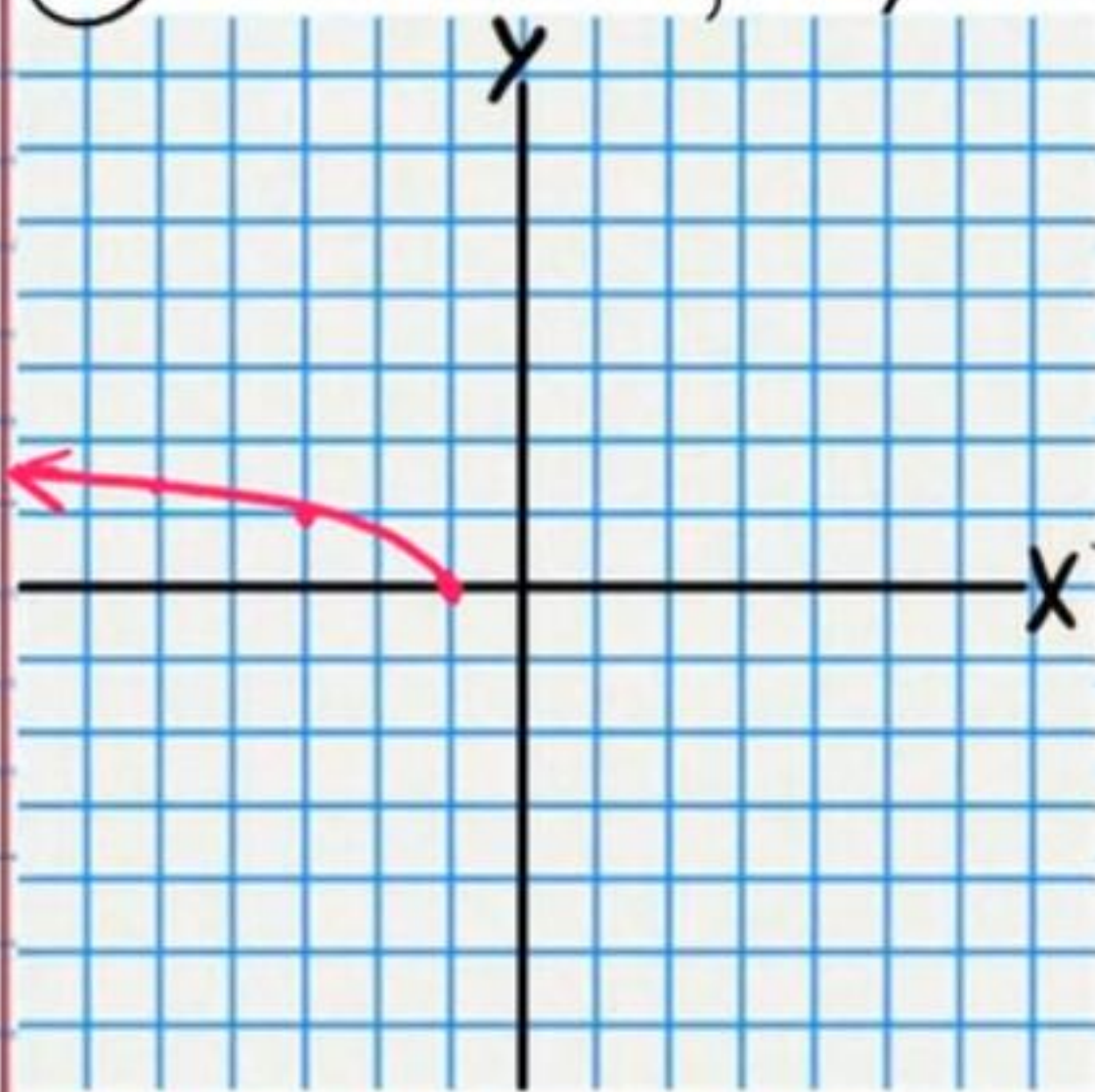
$\left(\frac{1}{4}, 4\frac{1}{4}\right)$

Eliminate the parameter

$x = t^2$, $y = t^2 + 4$

$y = x + 4$

21) $x = -2t - 1$, $y = \sqrt{t}$ Point #1 Point #2



$t = 0$

$x = -2(0) - 1$

$x = -1$

$y = \sqrt{0}$

$y = 0$

$(-1, 0)$

$t = 1$

$x = -2(1) - 1$

$x = -3$

$y = \sqrt{1}$

$y = 1$

$(-3, 1)$

Point #3

$t = 2$

$x = -2(2) - 1$ $y = \sqrt{2}$

$x = -5$

$(-5, \sqrt{2})$

Point #4

$t = 3$

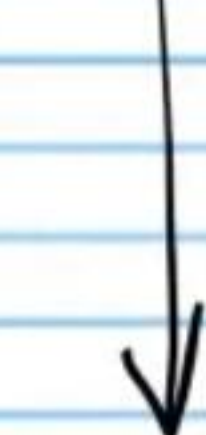
$x = -2(3) - 1$ $y = \sqrt{3}$

$x = -7$

$(-7, \sqrt{3})$

Eliminate the parameter

$x = -2t - 1$, $y = \sqrt{t}$



$x = -2y^2 - 1$



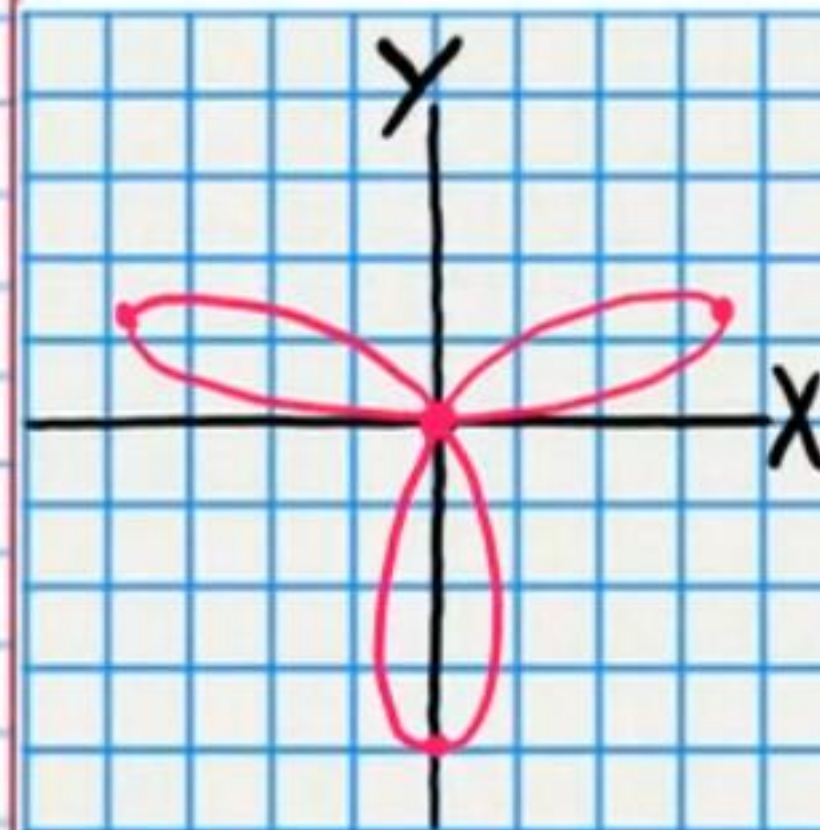
$y^2 = t$

Graph the following equations. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot.

22) $r = 4 \sin(3\theta)$

1st Point

2nd Point



$\theta = 0^\circ$

$r = 4 \sin(3(0^\circ))$

$r = 4 \sin 0^\circ$

$r = 4(0)$

$r = 0$

$(0, 0^\circ)$

$\theta = 30^\circ$

$r = 4 \sin(3(30^\circ))$

$r = 4 \sin 90^\circ$

$r = 4(1)$

$r = 4$

$(4, 30^\circ)$

3rd Point

4th Point

$\theta = -30^\circ$

$r = 4 \sin(3(-30^\circ))$

$r = 4 \sin(-90^\circ)$

$r = 4(-1)$

$r = -4$

$(-4, -30^\circ)$

$\theta = 90^\circ$

$r = 4 \sin(3(90^\circ))$

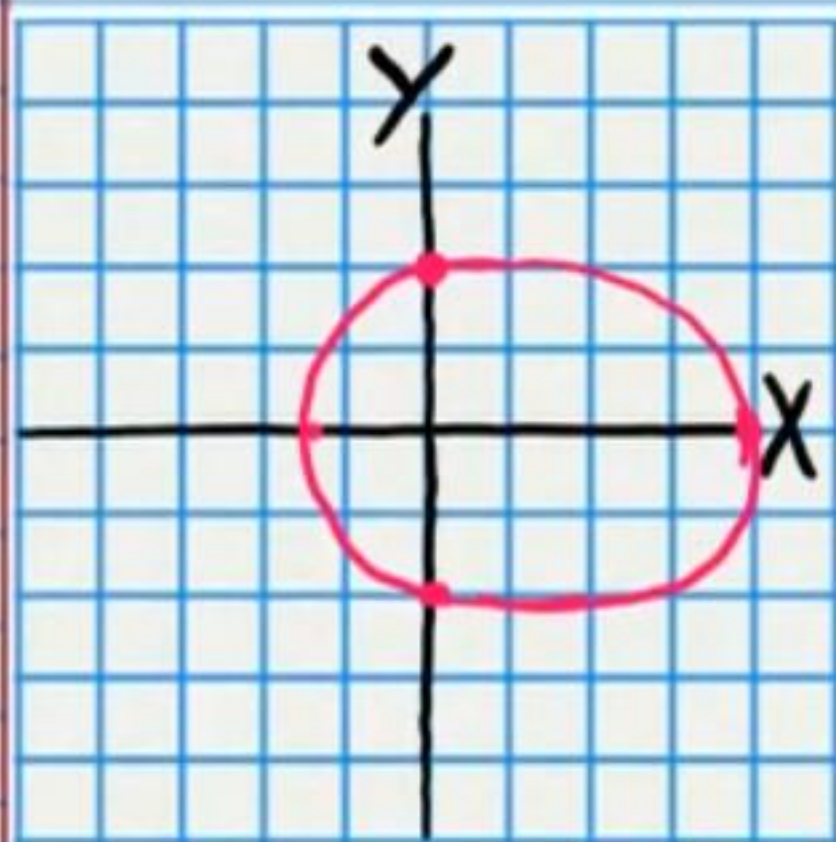
$r = 4 \sin 270^\circ$

$r = 4(-1)$

$r = -4$

$(-4, 90^\circ)$

$$(23) r = \frac{4}{2 - \cos \theta}$$



1st Point

$$\theta = 0^\circ$$

$$r = \frac{4}{2 - \cos 0^\circ}$$

$$r = \frac{4}{2 - 1}$$

$$r = 4$$

$$(4, 0^\circ)$$

2nd Point

$$\theta = 90^\circ$$

$$r = \frac{4}{2 - \cos 90^\circ}$$

$$r = \frac{4}{2 - 0}$$

$$r = 2$$

$$(2, 90^\circ)$$

3rd Point

$$\theta = -90^\circ$$

$$r = \frac{4}{2 - \cos(-90^\circ)}$$

$$r = \frac{4}{2 - 0}$$

$$r = 2$$

$$(2, -90^\circ)$$

4th Point

$$\theta = 180^\circ$$

$$r = \frac{4}{2 - \cos 180^\circ}$$

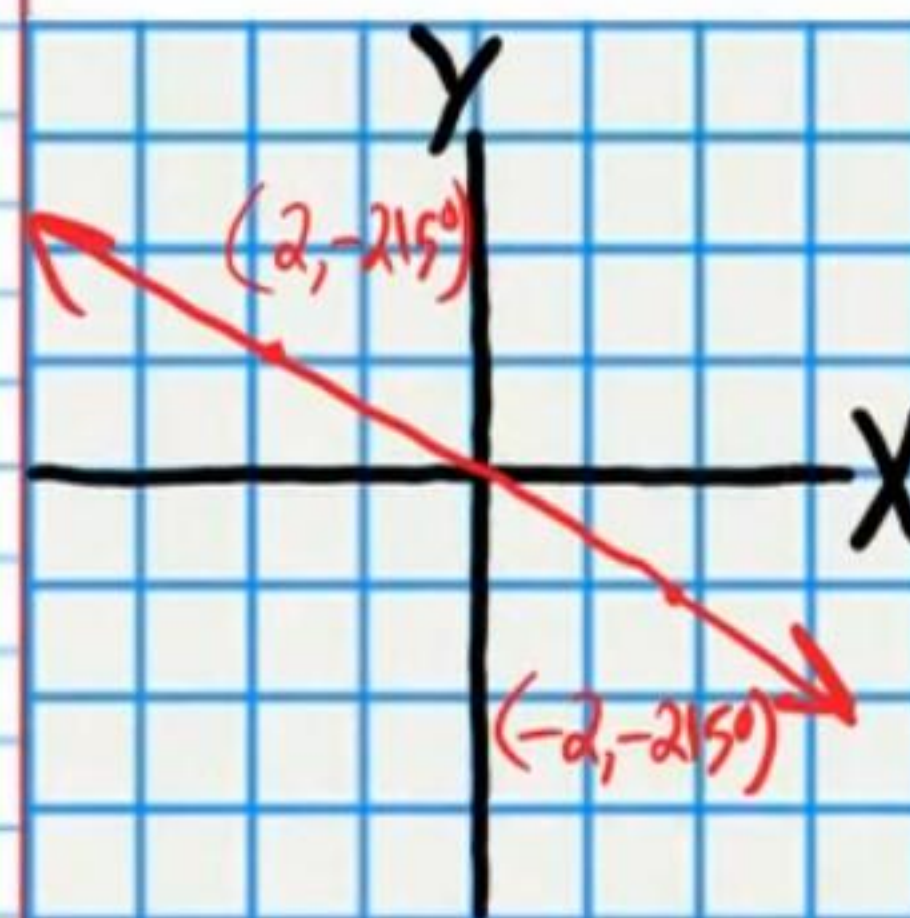
$$r = \frac{4}{2 - (-1)}$$

$$r = \frac{4}{3}$$

$$\left(\frac{4}{3}, 180^\circ\right)$$

$$(24) \theta = -215^\circ$$

* It is only necessary to write the coordinates of two points.



$$\tan \theta = \frac{y}{x}$$

$$\tan(-215^\circ) = \frac{y}{x}$$

$$-0.7 \approx \frac{y}{x}$$

Point #1

$$(2, -215^\circ)$$

Point #2

$$(-2, -215^\circ)$$

$$-0.7x \approx y$$

(25) Convert the equation below to polar form. Simplify if possible.

$$x^2 + y^2 = x + y$$

$$r^2 = r \cos \theta + r \sin \theta$$

$$r = \cos \theta + \sin \theta$$

$$\cos \theta = \frac{x}{r}$$

$$x = r \cos \theta$$

$$x^2 + y^2 = r^2$$

$$\sin \theta = \frac{y}{r}$$

$$y = r \sin \theta$$

